An Empirical Study of Vision-Language Model Fine-tuning for Radiology Report Generation:
Why Retrieval Baselines Win on Small-Scale Medical Benchmarks

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Abstract

Recent advances in general-purpose vision-language models (VLMs) such as Qwen2-VL have raised expectations that domain adaptation via parameter-efficient fine-tuning could match the performance of task-specific architectures on medical image-to-text tasks. We investigate this hypothesis on radiology report generation using the IU-Xray benchmark. Through a systematic comparison of seven approaches—including LoRA fine-tuning, classifier-guided prompting, visual feature fusion, text-only pre-training on MIMIC-CXR, and nearest-neighbor retrieval—we report three findings that, taken together, challenge a common assumption in this line of work.

First, all fine-tuning strategies plateau at BLEU-4 values between 6.6 and 9.0, with elaborate enhancement methods (CAT, VFF, MIMIC text pre-training) either matching or underperforming a no-hint LoRA baseline. Second, a trivial nearest-neighbor retrieval baseline requiring zero training achieves BLEU-4 = 9.08, exceeding every fine-tuned model. Third, clinical metrics reveal a different picture: our Visual Feature Fusion (VFF) variant achieves the highest CheXbert-14 micro-F1 (54.17) and RadGraph scores among fine-tuned methods despite a slightly lower BLEU-4, suggesting that lexical overlap metrics systematically undervalue visually-grounded improvements.

We attribute these patterns to three interacting factors: (1) the dominance of the language prior over the visual signal on a 2,069-sample training set, (2) gradient competition between language-side LoRA and visual-side adapters, and (3) the structural homogeneity of IU-Xray reports, which BLEU rewards through template matching rather than diagnostic accuracy. We argue that on small-scale medical benchmarks, BLEU is a poor proxy for vision-grounded generation quality, and that competitive retrieval baselines should be reported as standard practice.

1. Introduction

Automated radiology report generation has emerged as a benchmark task for evaluating multimodal models in the medical domain. The standard evaluation protocol on the IU-Xray dataset (Demner-Fushman et al., 2016) measures performance via lexical overlap metrics—BLEU, ROUGE-L, METEOR—and reports steady progress over the past five years, with current published BLEU-4 scores ranging from 16.5 (R2Gen) to over 19 (SERPENT-VLM).

The emergence of strong general-purpose vision-language models (VLMs) such as LLaVA, Qwen2-VL, and InternVL suggests an attractive alternative to task-specific architectures: fine-tune a pretrained VLM with parameter-efficient methods (e.g., LoRA) on a small medical dataset, optionally augmented with classifier-derived hints or domain pre-training. This direction is appealing because it reuses large-scale general pre-training and requires substantially less compute than training task-specific models from scratch. Several recent works have pursued variants of this strategy.

This work asks a different question. Rather than proposing a new method, we conduct a systematic empirical investigation of whether—and under what conditions—VLM fine-tuning on IU-Xray can yield reliable improvements over trivial baselines. We hold the base model (Qwen2-VL-7B) and dataset (IU-Xray, R2Gen standard split) fixed, and compare seven approaches spanning four design axes:

Pure fine-tuning: LoRA-based no-hint fine-tuning as a baseline.

Prompt-channel enhancement: Classifier-Aligned Training (CAT), which injects predicted disease hints into the language prompt.

Visual-channel enhancement: Visual Feature Fusion (VFF), which injects DenseNet classifier features into the vision token stream via gated cross-attention.

Domain pre-training: Text-only pre-training on 148K MIMIC-CXR reports prior to IU-Xray fine-tuning.

Reference points: A nearest-neighbor (NN) retrieval baseline that requires no training, and an Oracle setting where ground-truth disease keywords are leaked into the prompt as a sanity check.

Our findings are largely negative for fine-tuning. Every learned method either matches or underperforms the no-hint baseline on BLEU-4, while the NN retrieval baseline—a 30-second computation using DenseNet feature cosine similarity—outperforms all of them. Diagnostic experiments suggest that this is not a tuning artifact but a structural property of the task at this data scale: zero-shot Qwen2-VL generates the same "cardiomegaly is noted, with the heart occupying more than 50% of the thoracic cavity" template across visually distinct images, indicating that the language prior dominates the visual signal. Subsequent fine-tuning compresses this prior into IU-Xray's high-frequency templates ("the heart is normal in size... the lungs are clear") but does not produce genuinely image-conditioned generation.

A subtler finding emerges in the clinical metrics. While BLEU and ROUGE rank our visual-channel method (VFF) below the no-hint baseline, CheXbert and RadGraph metrics—which measure diagnostic entity overlap rather than lexical overlap—rank VFF as the strongest fine-tuned method. This divergence is consistent with the hypothesis that BLEU on IU-Xray primarily rewards template matching: a model that produces more diverse and clinically-grounded outputs may receive lower BLEU even when its outputs are more useful.

Contributions. We make three contributions:

(1) A controlled, head-to-head comparison of seven radiology report generation methods on a fixed setup, with both lexical and clinical metrics reported.

(2) Mechanistic analysis of why prompt-channel and visual-channel enhancements fail at this data scale, including evidence of gradient competition between LoRA and visual adapters (the VFF gate converges to 0.0327 regardless of learning rate schedule).

(3) An empirical case for treating the NN retrieval baseline as a mandatory reference point on small-scale medical generation benchmarks, and for supplementing BLEU with clinical metrics in this setting.

The remainder of the paper is structured as follows. Section 2 reviews related work on VLM-based medical report generation. Section 3 describes the experimental setup. Section 4 presents the seven methods. Section 5 reports results across lexical and clinical metrics. Section 6 analyzes failure mechanisms. Section 7 discusses implications and limitations.

2. Related Work

Task-specific architectures for radiology report generation. Early approaches frame report generation as a conditioned sequence-to-sequence task with a CNN encoder and a Transformer decoder. R2Gen (Chen et al., 2020) introduces a memory-driven Transformer with a relational memory module that records pattern information across the decoding process, achieving BLEU-4 = 16.5 on IU-Xray. R2GenCMN (Chen et al., 2021) extends this with cross-modal memory networks, reaching 17.0. These models are trained end-to-end on the target dataset and rely on architectural inductive biases tailored to radiology reports' patternized nature.

Retrieval as a strong baseline. A line of work predating recent VLM-based methods has observed that the highly templated nature of radiology reports makes simple retrieval competitive with learned generation. Liu et al. (2019) explicitly noted that a simple retrieval-based method could achieve comparative performance on this task, and Li et al. (2018) combined retrieval with template-based generation. R2Gen's own introduction acknowledges this finding as motivation for designing architectures that move beyond template matching. Despite this early evidence, retrieval baselines are rarely reported in recent VLM-based work, which we argue is a gap our study helps fill.

Medical VLMs via large-scale pre-training. A second line of work pre-trains general VLMs on large medical corpora before downstream fine-tuning. LLaVA-Med (Li et al., 2023) pre-trains on 213K MIMIC-CXR image-report pairs, with subsequent work (Kapadnis et al., 2024) reporting BLEU-4 = 18.6 for LLaVA-Med on IU-Xray under a unified evaluation protocol. SERPENT-VLM (Kapadnis et al., 2024) further introduces a self-refining alignment loss that maximizes similarity between pooled image representations and contextual representations of generated text, reaching BLEU-4 = 19.0. The common ingredient across these methods is access to large-scale medical pre-training data—typically the full MIMIC-CXR corpus (~377K images).

Parameter-efficient fine-tuning of general VLMs. A more recent direction asks whether the cost of medical pre-training can be avoided by fine-tuning general-purpose VLMs (Qwen2-VL, InternVL, LLaVA) with LoRA or similar techniques on the target medical dataset alone. Variants augment this with classifier-derived disease hints injected into the prompt, or with visual adapters that fuse external classifier features into the vision token stream. The implicit assumption is that strong general pre-training plus a small amount of domain adaptation can compensate for the absence of medical pre-training data.

Our position. This work directly tests that assumption on IU-Xray and finds it does not hold at this data scale. We further extend the existing evidence on retrieval baselines (Liu et al., 2019) into the modern VLM era, showing that the gap between learned generation and trivial retrieval has not closed—and is in fact amplified—under Qwen2-VL + LoRA. Finally, we argue that the standard BLEU/ROUGE evaluation protocol, when applied to a benchmark with highly repetitive ground-truth reports, systematically rewards template matching over genuine visual grounding, and that supplementing lexical metrics with clinical metrics (CheXbert, RadGraph) reveals a partially different story.

3. Experimental Setup

3.1 Task and Dataset

We adopt the standard radiology report generation formulation: given a pair of chest X-ray views (frontal and lateral), generate the free-text "findings" section of the corresponding radiology report. We use the IU-Xray dataset (Demner-Fushman et al., 2016) with the R2Gen standard split: 2,069 training samples, 296 validation, 590 test. Each sample consists of dual-view X-ray images paired with a findings paragraph typically 10–50 tokens in length.

A defining characteristic of IU-Xray is its lexical homogeneity: 92.6% of training reports contain at least one of the normal-finding keywords normal, clear, or unremarkable. This skews the learning signal toward a small set of high-frequency templates, a property we revisit when analyzing model behavior in Section 6.

3.2 Base Model

All learned methods build on Qwen2-VL-7B-Instruct as the base VLM. We apply LoRA fine-tuning with rank 16, alpha 32, and target modules {q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj}. Training uses bfloat16 precision on a single NVIDIA A100 (40GB or 80GB depending on memory requirements). A representative fine-tuning run takes approximately 11 minutes per epoch on IU-Xray; the MIMIC text pre-training stage (Section 4.4) requires approximately 7.5 hours per epoch on 148K reports.

3.3 Auxiliary Classifier

Several methods rely on disease classification as an auxiliary signal. We use DenseNet-121, initialized from the torchxrayvision weights pre-trained on multiple chest X-ray corpora, and fine-tuned on IU-Xray for 10-class multi-label classification. The classifier achieves a macro-F1 of 0.221 on the test set with per-class optimal thresholds, reflecting the noise level inherent in this auxiliary signal—a factor that becomes relevant when interpreting the failure of classifier-hint methods.

3.4 Evaluation Metrics

We report two families of metrics:

Lexical metrics: BLEU-1/2/3/4, ROUGE-1/2/L, METEOR. These measure n-gram and lexical overlap between generated and reference reports and are the de facto standard for radiology report generation benchmarks.

Clinical metrics: CheXbert-5 and CheXbert-14 F1 scores (micro and macro), measuring agreement on the 5 most clinically critical and 14 standard CheXpert disease labels respectively, extracted from generated and reference text via the CheXbert labeler. RadGraph F1 (Simple and Partial) measures overlap of clinical entities and their relations as parsed by the RadGraph schema.

We additionally report the number of **unique outputs** across the 590 test samples, which serves as a coarse diagnostic for mode collapse: a model producing 590 distinct outputs is fully diverse, while a model producing very few distinct outputs has collapsed to a small set of templates.

3.5 Decoding and Reproducibility

All learned models use beam search with beam size 4. Generation length is capped at 100 tokens. Each reported number is from a single run on the full 590-sample test set; we do not report variance across seeds, which is a limitation acknowledged in Section 7.

4. Methods

We describe the seven methods compared in this study. Sections 4.1–4.5 present the five learned approaches; Sections 4.6 and 4.7 describe the two reference baselines.

4.1 LoRA Baseline (No-Hint Fine-tuning)

Our baseline applies LoRA fine-tuning to Qwen2-VL-7B-Instruct on the IU-Xray training set with no auxiliary signal. The input is the concatenation of the two-view chest X-ray and a standard instruction prompt requesting a findings paragraph; the target is the ground-truth findings text. We use cross-entropy loss with teacher forcing during training and beam search (beam size 4) at inference. This setup serves as the reference point against which all enhancement methods are compared.

4.2 CAT: Classifier-Aligned Training

Motivation. A natural way to inject domain knowledge into a general-purpose VLM is through the prompt: if the model is told which abnormalities are likely present in the image, it should produce a more accurate report. Earlier experiments revealed that training the model with ground-truth-extracted hints but providing classifier-predicted hints at inference produces a distribution mismatch: the BLEU-4 collapses to 8.69, essentially matching the no-hint baseline. CAT addresses this by using the same DenseNet-121 classifier to generate hints during both training and inference, ensuring strict train-test consistency.

Implementation. For each image, the DenseNet-121 classifier predicts probabilities over 10 pathology labels. Labels with predicted probability above an optimized per-class threshold are concatenated into a comma-separated string and inserted into the prompt as "Findings to consider: {label1, label2, ...}". The same procedure is used in training and inference.

Expected behavior. Under the standard hypothesis that hints provide useful prior information, CAT should match or exceed the no-hint baseline. We report the actual outcome in Section 5.

4.3 VFF: Visual Feature Fusion

Motivation. The prompt channel forces auxiliary information through a discrete tokenization bottleneck and competes with the natural language instruction. If the prompt-based hint signal fails (a possibility we observe with CAT in Section 5), an alternative is to inject auxiliary information directly into the visual token stream, bypassing the language channel entirely.

Implementation. We extract the penultimate-layer features (1024-d) from the fine-tuned DenseNet-121 classifier for each input image. A gated cross-attention module fuses these features with the output of Qwen2-VL's vision merger (the projection from ViT tokens to the language model's embedding space). The fused output replaces the original vision tokens before being passed to the language model.

The gate is parameterized as $g = \tanh(w)$ with w initialized to zero, so the gate value starts at exactly 0 and the model begins training identical to the no-hint baseline. This design provides what we call a **lower-bound guarantee**: even if the VFF adapter fails to learn anything useful, the gate remains near zero and the model recovers baseline behavior. The gate, adapter parameters, and language-side LoRA are trained jointly with the standard cross-entropy loss.

Expected behavior. If the visual signal from DenseNet provides complementary information beyond what Qwen2-VL's own vision encoder extracts, the gate should grow and BLEU-4 should improve. The actual gate trajectory and outcome are reported in Section 5.

4.4 MIMIC-CXR Text-Only Pre-training

Motivation. LLaVA-Med achieves BLEU-4 = 18.6 on IU-Xray largely through pre-training on 213K MIMIC-CXR image-report pairs. Replicating this is infeasible under our hardware constraints (the full MIMIC-CXR image archive exceeds 130GB and requires credentialed access). We instead ask whether text-only pre-training on MIMIC-CXR reports—which are far smaller in disk footprint and substantially easier to obtain—can capture useful linguistic patterns of radiology reports before downstream visual fine-tuning.

Implementation. We collected 148K MIMIC-CXR findings paragraphs and constructed three text-to-text pre-training tasks: (1) generate the impression section given the findings section, (2) continue a truncated findings paragraph given its first half, and (3) generate a full findings paragraph given a disease-keyword hint. We apply LoRA to Qwen2-VL's language modules and train for two epochs on the combined task mixture (validation loss decreases from 0.94 to 0.88). The image modality is unused at this stage. We then fine-tune the resulting checkpoint on IU-Xray following the LoRA Baseline procedure.

Expected behavior. If radiology-specific linguistic priors transfer to IU-Xray, this method should outperform the no-hint baseline. We report the actual outcome in Section 5.

4.5 Disease-Hint-Guided Prompting

Configuration. This method is the LoRA Baseline configured with the disease-hint prompting strategy described in our preliminary experiments: training-time hints are extracted from the ground-truth report (via keyword matching against the 10-label vocabulary), inference-time hints come from the DenseNet-121 classifier, 30% hint dropout is applied during training, abnormal cases are oversampled 2×, and prompt phrasings are randomized to prevent the model from over-relying on hint format. This was our primary configuration for several iterations before the systematic comparison described in this paper revealed that classifier-aligned and visual-feature variants did not improve upon it.

We include results for this configuration where available; some clinical metrics were not computed under this setting due to compute constraints and are marked accordingly in Section 5.

4.6 NN Retrieval (Zero-Training Reference)

Motivation. Following the early observation by Liu et al. (2019) that retrieval can be competitive for radiology report generation, and given the templated nature of IU-Xray ground-truth reports, we include a trivial nearest-neighbor retrieval baseline to bound the contribution of “template matching” in any reported result.

Implementation. For each test image, we extract penultimate-layer features (1024-d) from the fine-tuned DenseNet-121 classifier and find the training image with the highest cosine similarity. The ground-truth report of that training image is returned as the prediction—without modification, without any generation step, and without any training of the prediction module. The entire pipeline runs in approximately 30 seconds on the full test set.

We emphasize that this baseline involves zero learnable parameters in the generation step; the only "learning" is the prior fine-tuning of the DenseNet classifier on the same training set.

4.7 Oracle (Ground-Truth Hint Leak)

Motivation. To distinguish "hints help" from "hints leak ground-truth information," we run an Oracle configuration in which the disease-keyword hint is extracted from the test sample's ground-truth report and injected into the prompt. This is a deliberately leaky setup used only as a sanity check.

Interpretation. Oracle BLEU-4 is not a meaningful upper bound on vision-grounded generation: a hint like "cardiomegaly" extracted from the ground-truth essentially reduces the task to "produce a report containing the word 'cardiomegaly'," which any reasonable language model can satisfy by template substitution rather than visual reasoning. We report Oracle results for diagnostic value but explicitly argue against treating it as a method comparison target.

5. Results

5.1 Main Results

Tables 1a and 1b report the seven methods on the full 590-sample IU-Xray test set across lexical and clinical metrics, respectively.

Table 1a. Lexical metrics on the IU-Xray test set (590 samples).

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE-1	ROUGE-2	ROUGE-L	METEOR	Unique
Fine-tuning methods									
LoRA Baseline	29.97	17.71	11.85	8.48	41.26	15.24	30.41	23.95	30
Disease-Hint	—	—	—	~9.0	—	—	—	—	—
CAT	22.14	13.27	9.20	6.63	36.84	14.31	27.50	20.19	29
VFF	27.06	16.54	11.47	8.39	40.15	16.04	29.81	23.23	43
MIMIC Pre-training	22.46	13.60	9.38	6.80	34.27	13.63	27.15	20.02	17
Retrieval / Reference									
NN Retrieval (zero training)	29.83	18.11	12.38	9.08	37.90	14.82	27.62	—	—
Oracle (GT leak)	37.54	25.83	19.86	16.14	47.47	22.80	35.87	32.28	—

Table 1b. Clinical metrics on the IU-Xray test set (590 samples).

Method	CheXbert-5		CheXbert-14		RadGraph	
	micro-F1	**macro-F1**	**micro-F1**	**macro-F1**	**Simple**	**Partial**
Fine-tuning methods						
LoRA Baseline	0.00	0.00	51.63	5.26	35.75	33.15
Disease-Hint	—	—	—	—	—	—
CAT	3.10	1.19	53.66	5.39	34.37	32.87
VFF	0.00	0.00	54.17	5.02	37.11	34.36
MIMIC Pre-training	10.84	2.67	36.35	4.95	32.57	30.79
Retrieval / Reference						
NN Retrieval (zero training)	25.33	17.73	41.45	15.47	33.06	30.29
Oracle (GT leak)	23.78	11.11	54.51	12.77	43.08	40.19

We organize the discussion around three observations.

Observation 1: All fine-tuning methods plateau in a narrow band, and elaborate enhancements underperform the no-hint baseline. The LoRA Baseline achieves BLEU-4 = 8.48. CAT (classifier-aligned prompt hints) drops to 6.63, a 1.85-point decrease. VFF (visual feature fusion) reaches 8.39, statistically indistinguishable from baseline. MIMIC text pre-training results in 6.80, the worst among learned methods. The Disease-Hint configuration achieves approximately 9.0 on BLEU-4 (other metrics not computed under this setting). Across all five learned methods, BLEU-4 spans only the range 6.63–9.0, despite substantial architectural and training-pipeline differences. We interpret this as evidence that fine-tuning has hit a structural ceiling at this data scale rather than failing for method-specific reasons.

Observation 2: A zero-training retrieval baseline exceeds every fine-tuned method on BLEU. The NN Retrieval baseline achieves BLEU-4 = 9.08, exceeding the LoRA Baseline (8.48), VFF (8.39), CAT (6.63), and MIMIC pre-training (6.80). On BLEU-1 (29.83), BLEU-2 (18.11), and BLEU-3 (12.38), retrieval is the highest-scoring method overall, exceeding even Disease-Hint where comparable numbers are available. This replicates and strengthens the observation of Liu et al. (2019) in the modern VLM setting: the gap between learned generation and trivial retrieval has not closed despite five years of architectural progress on this benchmark.

Observation 3: Clinical metrics tell a different story than BLEU. On CheXbert and RadGraph metrics, the ranking of methods diverges sharply from the BLEU ranking:

> **VFF achieves the best CheXbert-14 micro-F1 among all fine-tuned methods (54.17)**, exceeding the LoRA Baseline (51.63) and CAT (53.66). VFF also achieves the highest RadGraph-Simple (37.11) and RadGraph-Partial (34.36) scores in the entire table.
>
> **NN Retrieval dominates CheXbert-5**, with micro-F1 = 25.33 and macro-F1 = 17.73, compared to 0.00 for both the LoRA Baseline and VFF. CheXbert-14 macro-F1 also strongly favors retrieval (15.47 vs. 5.26 for baseline).

> The LoRA Baseline's **CheXbert-5 micro-F1 of 0.00** is a striking diagnostic signal: despite producing fluent radiology-style text and achieving competitive BLEU, the fine-tuned model produces essentially no correct clinical entities on the five most clinically critical CheXpert categories.

The divergence between lexical and clinical metrics is consistent with the hypothesis that BLEU on IU-Xray primarily rewards lexical patterns associated with the dataset's high-frequency “normal” templates, while clinical metrics measure whether the generated text contains the correct diagnostic entities. VFF improves clinical entity accuracy but disrupts the templated phrasing that BLEU rewards; the LoRA Baseline produces fluent templates but very few correct clinical entities for the most critical labels.

5.2 Output Diversity

Across the 590 test samples, the LoRA Baseline produces only 30 unique outputs, indicating heavy mode collapse. VFF produces 43 unique outputs—the highest among fine-tuning methods—consistent with its stronger clinical performance: the model is generating more image-specific content rather than collapsing to a small set of templates. MIMIC pre-training produces only 17 unique outputs, the lowest in the table, confirming a severe template-collapse failure mode (we analyze this in Section 6.4).

5.3 Gap to Reported SOTA

Our LoRA Baseline (BLEU-4 = 8.48) is substantially below published numbers on this benchmark: R2Gen at 16.5 (Chen et al., 2020), R2GenCMN at 17.0 (Chen et al., 2021), LLaVA-Med at 18.6 (Kapadnis et al., 2024), and SERPENT-VLM at 19.0 (Kapadnis et al., 2024). We attribute this gap primarily to the absence of large-scale medical pre-training—LLaVA-Med uses 213K MIMIC-CXR pairs—which our hardware-constrained text-only substitute (Section 4.4) does not adequately replace. We discuss what this gap implies, and what it does not imply, in Section 7.

6. Failure Mechanism Analysis

The numerical patterns in Section 5 raise a natural question: are these methods failing because of implementation issues, or because of structural properties of the task at this data scale? We argue for the latter, based on five lines of diagnostic evidence.

6.1 The Language Prior Dominates the Visual Signal

We ran zero-shot Qwen2-VL-7B (no LoRA, no fine-tuning) with few-shot prompting on a 100-sample subset of the IU-Xray test set. BLEU-4 was 1.67, as expected for an untuned general VLM. The diagnostic finding, however, is not in the metric but in the outputs:

“Cardiomegaly is noted, with the heart occupying more than 50% of the thoracic cavity.”

This exact sentence appeared as part of the output in 4 out of 5 randomly inspected samples. **All four had ground-truth reports describing normal cardiac findings.** The model was not looking at the image; it was emitting a high-likelihood medical template anchored on the prompt's “chest X-ray” framing.

This is the central diagnostic for the entire study. Subsequent LoRA fine-tuning compresses this prior into

IU-Xray's dominant templates ("the heart is normal in size... the lungs are clear") rather than producing genuinely image-conditioned generation. The 0.00 CheXbert-5 micro-F1 of the LoRA Baseline (Section 5.1) is the direct numerical signature of this mechanism: the model produces fluent, BLEU-competitive text that contains essentially no correct clinical entities on the most critical labels.

6.2 Why CAT Fails: Noisy Hints and Shortcut Learning

CAT's failure (BLEU-4 = 6.63, 1.85 points below baseline) has three contributing factors, identified through targeted analyses.

Hint noise. The DenseNet classifier achieves macro-F1 = 0.221 on IU-Xray (Section 3.3). Inspecting individual predictions, only approximately 55% of predicted hint tokens match the ground-truth disease label set for the same image. The model is being trained on systematically noisy auxiliary input.

Capacity to learn noisy mappings. With 2,069 training samples and a noisy hint signal, LoRA fine-tuning cannot reliably learn the conditional mapping "noisy hint set → correct disease description." Each individual hint–phrase association may be witnessed only a handful of times in training, well below what we would expect to be sufficient for a 7B-parameter model to commit to it.

Shortcut learning. We ran a conditional generation diagnostic: for the same image, we generated outputs while varying the injected hint set across plausible alternatives (e.g., "cardiomegaly" vs. "atelectasis" vs. "effusion"). **The generated output was essentially identical across hint variants.** The model has learned to use the presence of a hint as a signal that "this is a short-template prediction context," but ignores the hint's actual content. This is consistent with prior observations on conditional generation models trained on noisy auxiliary inputs.

6.3 Why VFF Plateaus: Gradient Competition

VFF reaches BLEU-4 = 8.39, statistically indistinguishable from the no-hint baseline, despite achieving the best clinical metrics. The mechanistic explanation is in the gate dynamics.

The gate value converges to 0.0327. Recall (Section 4.3) that the gate is parameterized as $g = \tanh(w)$ with w initialized to zero, so it starts at exactly 0 and grows only if the visual fusion signal proves useful to the cross-entropy loss. We tried multiple interventions: increasing the gate's learning rate to 10× and 100× the adapter learning rate, decoupling the gate optimizer from the rest of the model, and warming up the adapter before allowing the gate to move. **The gate's converged value remained in the range 0.03–0.05 across all configurations.**

We interpret this as evidence of gradient competition. The cross-entropy objective can be driven down through two pathways: improving visual grounding (the VFF pathway) or compressing the language model into the high-frequency templates of the training distribution (the LoRA pathway). On 2,069 samples, the second pathway is far easier to optimize—the gradient signal for "match the training templates" is dense and consistent, while the gradient signal for "use the auxiliary visual feature to disambiguate cases" is sparse and noisy. The LoRA pathway absorbs the optimization budget that should flow through VFF.

The clinical metric result (Section 5.1) is the residual evidence that VFF is *not* failing: even with the gate near zero, the small amount of visual signal that does flow through is enough to improve CheXbert-14 and RadGraph scores. The method is not broken; it is being out-competed by an easier optimization pathway that happens to also receive higher reward from BLEU.

6.4 Why MIMIC Pre-training Backfires: Template Memorization

Text-only pre-training on 148K MIMIC-CXR findings paragraphs reduces BLEU-4 from 8.48 to 6.80—worse than no pre-training at all. The number of unique outputs across the 590-sample test set drops to 17, the lowest in our entire experiment. Inspecting the generations:

Multiple distinct test images—including images with clear pathology in their ground-truth reports—produced the exact same output:

> *"The heart is normal in size. The mediastinum is unremarkable. The lungs are clear."*

This phrase is one of the highest-frequency templates in MIMIC-CXR. The model over-memorized it during the text pre-training stage, and downstream visual fine-tuning on 2,069 IU-Xray samples could not displace this prior. Attempts to mitigate the collapse via abnormal-case oversampling at the fine-tuning stage reduced the rate of template emission but did not recover BLEU.

The failure mode is consistent with the language-prior diagnosis in Section 6.1, but amplified: we have effectively given the model *more* template prior before exposing it to the small visual training set, making the visual signal even less competitive at update time.

6.5 Why Oracle = 16.14 Is Not a Real Upper Bound

The Oracle configuration (Section 4.7) achieves BLEU-4 = 16.14, ROUGE-L = 35.87, and METEOR = 32.28—numbers that, if interpreted as an upper bound, would suggest substantial headroom above our fine-tuned methods. We argue this interpretation is incorrect.

Oracle hints are extracted from the ground-truth report by keyword matching. If the ground-truth contains "cardiomegaly," the prompt instructs the model to consider cardiomegaly. The task is no longer "look at the image and generate a report"; it is "produce a report that includes the words I just told you." A language model with no visual capability whatsoever could perform this task by template-substituting hint keywords into a generic radiology phrasing.

The clinical metrics tell the same story from the other direction: Oracle's CheXbert-14 micro-F1 (54.51) is only marginally higher than VFF's (54.17) despite Oracle having ground-truth keywords leaked into its input. If Oracle reflected true vision-grounded capability, we would expect a much larger clinical gap.

A more defensible upper bound for honest, non-cheating methods on this benchmark and setup is probably in the BLEU-4 = 10–12 range. This is consistent with the gap between our methods (8–9) and published task-specific architectures with substantially more inductive bias (16–17).

7. Discussion

7.1 What This Study Does and Does Not Show

What it shows. On IU-Xray at the 2,069-sample training scale, LoRA fine-tuning of Qwen2-VL-7B with prompt-channel or visual-channel enhancements does not improve upon a no-hint baseline, and is exceeded on BLEU by a zero-training nearest-neighbor retrieval baseline. The same retrieval baseline dominates on CheXbert-5 clinical metrics. Within the fine-tuned methods, the clinical metrics rank methods differently than BLEU, with VFF the strongest fine-tuned method on CheXbert-14 and RadGraph despite a slightly lower BLEU than baseline.

What it does not show. Our results do not establish that general-purpose VLM fine-tuning is ineffective for medical report generation in general. With access to large-scale medical pre-training (e.g., 213K MIMIC-CXR pairs as in LLaVA-Med), the situation is documented to be substantially different. Our claims are scoped to the data regime where domain-specific pre-training is unavailable and the target dataset is small (~2K samples).

Our results also do not establish that retrieval is preferable to generation in clinical deployment. Retrieval returns the *exact* ground-truth report of another patient, which is unsuitable for production use; its value here is as a diagnostic baseline that bounds the BLEU contribution attributable to template matching.

7.2 Implications for Benchmark Design

The most actionable implication of our findings concerns how the radiology report generation community evaluates progress on IU-Xray-scale benchmarks. We make three suggestions:

(1) Report NN retrieval as a standard baseline. A method that does not exceed nearest-neighbor retrieval on a given benchmark should not be reported as a positive result on that benchmark, regardless of architectural novelty. This is computationally trivial (30 seconds on IU-Xray) and provides a meaningful lower bar.

(2) Report clinical metrics alongside lexical metrics. Our VFF result—the best clinical metrics paired with mediocre BLEU—shows that the two metric families can diverge, and that BLEU alone may systematically undervalue genuine visual grounding on templated benchmarks.

(3) Be cautious with "Oracle" or "upper bound" framings that leak ground-truth signal. The numerical gap to Oracle is not in general a measure of how much room remains for improvement.

7.3 Limitations

Several limitations of this study should be noted.

Single-seed evaluation. Each reported number is from a single training run. We did not have the compute budget to estimate variance across random seeds, which is a known weakness for negative-results papers in particular: a failure to improve on baseline could in principle reflect bad seed luck rather than a structural limitation. The consistency of the pattern across five distinct methods makes this less likely but does not eliminate the concern.

Single base model. All learned methods build on Qwen2-VL-7B. We do not test whether the failure pattern is specific to this model or generalizes to other current VLMs (LLaVA, InternVL, Pixtral).

Single dataset. Our claims are specific to IU-Xray. MIMIC-CXR (377K images) sits in a substantially different data regime and our findings should not be extrapolated to it.

No human evaluation. All metrics are automated. Human radiologist evaluation of the generated reports—particularly comparing fluent-but-clinically-empty outputs from the LoRA Baseline against more clinically grounded VFF outputs—would substantially strengthen the case for our interpretation of the clinical-vs-lexical divergence.

Disease-Hint partial reporting. The Disease-Hint configuration (Section 4.5) was our primary configuration for an extended period but does not have complete clinical metrics in our final table. This is a presentation gap rather than a finding gap, but should be noted.

7.4 Future Directions

Three directions follow naturally from these results.

Retrieval-augmented generation (RAG). Given that pure retrieval achieves BLEU-4 = 9.08 and the strongest CheXbert-5 scores, while fine-tuning achieves better clinical entity F1 on the 14-label setup, a natural next step is to combine the two: retrieve a candidate report from the training set, then fine-tune the VLM to *edit* the retrieved report conditional on the input image. This is a substantially different objective than "generate from scratch" and avoids the template-collapse failure mode.

Cleaner evaluation tasks. Visual Question Answering (VQA) and anatomy grounding provide cleaner evaluation signals than free-text generation: the metric directly measures what we want the model to do, with no template-matching confound. Migrating to these tasks would allow sharper measurement of whether VLM fine-tuning is providing genuine visual capability.

Pre-trained medical VLMs as the base. Replacing Qwen2-VL with LLaVA-Med or MAIRA-2 as the starting checkpoint—rather than trying to substitute for medical pre-training via text-only methods (Section 4.4)—is the most direct path to higher absolute numbers. The findings of this study would still apply as a diagnostic framework but the base capability would be substantially higher.

8. Conclusion

We conducted a controlled empirical study of vision-language model fine-tuning for radiology report generation on IU-Xray, comparing seven methods spanning prompt-channel enhancement, visual-channel enhancement, domain pre-training, and zero-training retrieval. All learned methods plateau in the BLEU-4 = 6.6–9.0 range; a 30-second nearest-neighbor retrieval baseline achieves BLEU-4 = 9.08, exceeding every fine-tuned model. Diagnostic experiments trace this pattern to the dominance of the language prior on a 2,069-sample training set, gradient competition between language-side LoRA and visual-side adapters, and the structural homogeneity of IU-Xray reports that BLEU rewards through template matching.

A subtler finding emerges from the clinical metrics: VFF, which underperforms baseline on BLEU, achieves the strongest CheXbert-14 and RadGraph scores among fine-tuned methods, suggesting that lexical overlap metrics systematically undervalue genuine visual grounding on this benchmark. Taken together, we argue that retrieval baselines should be reported as standard practice, that clinical metrics should accompany lexical metrics, and that the gap between learned generation and trivial retrieval—first noted by Liu et al. in 2019—has not closed despite five years of architectural progress on this benchmark.

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